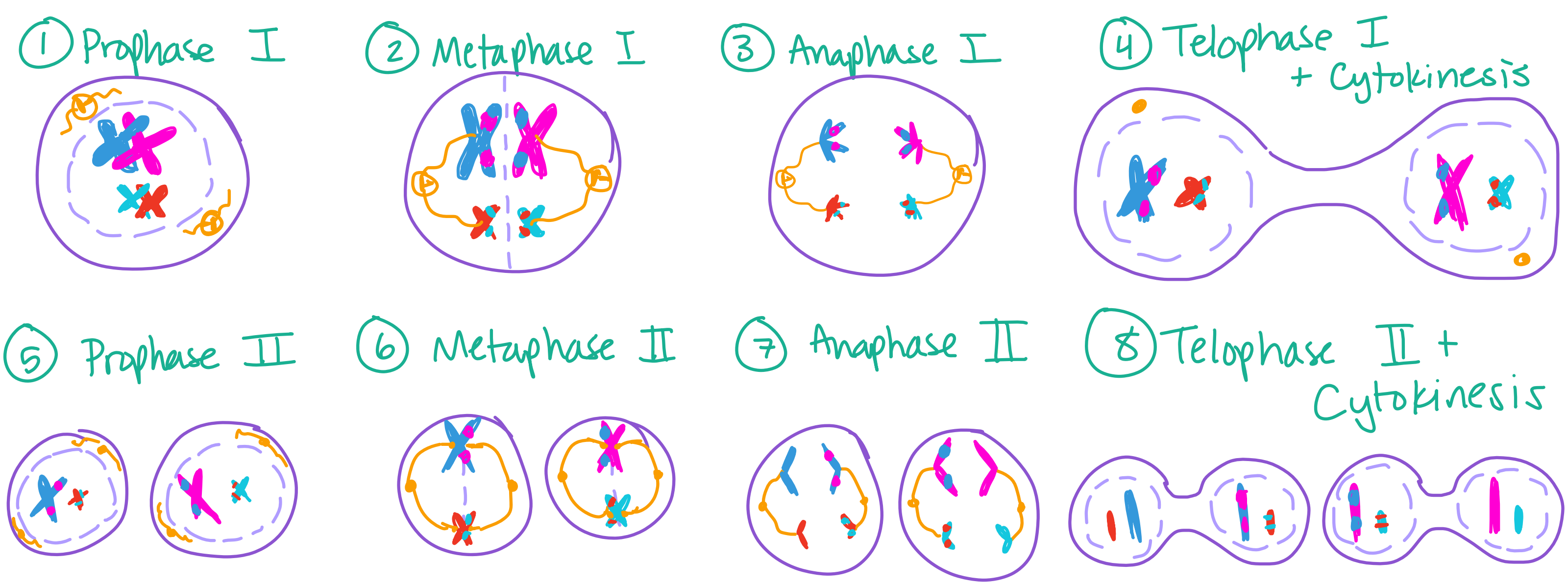
<u>unit 5</u>

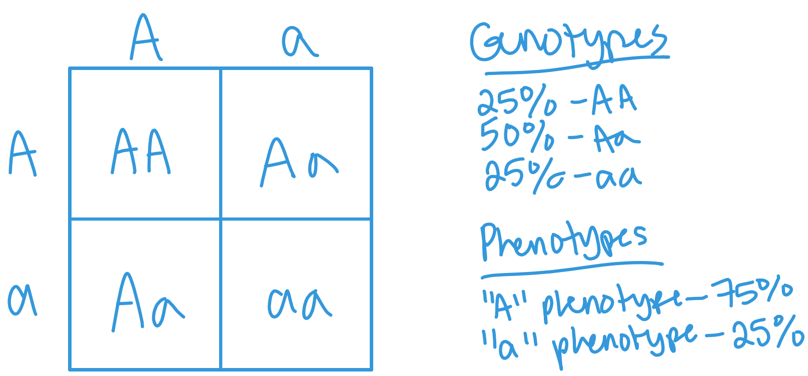
- **Diploid—**two sets of chromosomes (represented by "2n"), somatic cells
  - *** Human somatic cells have 46 total chromosomes, and human gametes have 23 chromosomes.
- **Haploid—**one set of chromosomes (represented by "n"), gametes
- **Sperm—**male reproductive gamete
- **Egg—**female reproductive gamete
- **Sexual Reproduction—**randomized process of two parents contributing DNA for increased genetic diversity
- **Fertilization—**sperm fuses with egg to create a zygote
- **Zygote—**diploid cell that contains the genetic information from two parents
- **Meiosis—**a specific cell division process for the purpose of making genetically unique gametes
- **Gene—**unit of heredity, encodes for a specific protein which creates a certain phenotype
- **Trait—**physical expression of a gene
- **Locus (loci; plural)—**the specific location on a chromosome where a specific gene is located
- **Homologous Chromosomes (Homologues)—**chromosomes that are the same size and have the same loci (same genes that control the same phenotypes), found in diploid cells
  - *** Homologues are duplicated before cell division but do not separate at the centromere; rather, chromosomes themselves are separated during meiosis.
- **Karyotype—**ordered display of pairs of homologous chromosomes by size
- **Sex chromosomes—**determine an organism's sex
  - **XX Chromosomes—**human female
  - **XY Chromosomes—**human male
- **Autosomes—**chromosomes that do not determine an organism's sex

- **Meiosis—**creating gametes from a somatic cell for sexual reproduction and greater increased genetic diversity through crossing over and independent assortment.
  - **Meiosis I—**Crossing over occurs and homologous chromosomes separate. The cell divides once. Ends with two genetically different haploid daughter cells.
    - **Prophase I—**Duplicated homologues pair up and exchange genetic information at certain loci and form a tetrad. Then, the nuclear membrane disappears and the meiotic spindle begins to form.
      - **Tetrad—**a group of two homologues; four chromatids
      - **Crossing Over—**exchanging information at the same loci (same genes, same physical characteristics) between homologues for increased genetic diversity. Can occur at any loci.
      - **Chiasmata—**x-shaped region on both chromosomes where crossing over occurs
      - **Meiotic spindle—**pulls sister chromatids and homologues to opposite ends of cell to divide info

    - **Metaphase I**—tetrads line up on the metaphase plate. Independent assortment occurs. The meiotic spindle attaches to a kinetochore of each homologues from each pair.
        - **Independent Assortment**—the random line-up of chromosomes that increases genetic diversity at the metaphase plate during Metaphase I; determines which chromosomes (one from mom, one from dad) go to which gametes
        - **Kinetochore**—across from the centromere in sister chromatids
    - **Anaphase I**—one homologue from each pair is pulled to one end of the cell by the meiotic spindle, and each other homologue is pulled to the other end, still attached via Kinetochore
    - **Telophase I**—meiotic spindle degrades, nuclear membrane re-forms. No chromosome replication. Sister chromatids still attached at the Kinetochore.
    - **Cytokinesis**—division of the cytoplasm
        - **Cleavage furrow (animals)**—separates the cytoplasm
        - **Cell plate (plants)**—separates the cytoplasm and becomes part of the daughter cells' cell walls
- **Meiosis II**—starts with two genetically different daughter cells. Sister chromatids will separate. Results in four haploid genetically diverse daughter cells.
    - **Prophase II**—nuclear membranes disappear. Meiotic spindle re-forms.
    - **Metaphase II**—chromosomes line up along the metaphase plate, each sister chromatid is attached via kinetochore to the meiotic spindle.
    - **Anaphase II**—sister chromatids pulled to opposite poles of the cell
    - **Telophase II**—chromatids uncoil, nuclear membrane re-forms.
    - **Cytokinesis**—cytoplasm divides



- **Mendelian Genetics—**simple two-allele system discovered by Gregor Mendel—dominant trumps recessive
- **True-breeding—**when self-pollinated, this plant always produces the same desired phenotype
- **Hybridization—**breeding two true-breeding plants of different phenotypes
- **P generation—**two true-breeding plants of different phenotypes
- **F1 generation—**hybrid offspring of P generation
- **F2 generation—**offspring of F1 generation, always seemed to display a certain trait in a 3:1 ratio
- **Allele—**certain version of a gene that encodes for a specific phenotype, determined by its set of nucleotides. A child always inherits just two alleles, or two copies of a gene—one from mom, one from dad.
- **Dominant allele—**the allele that is completely displayed in a hybrid (capital letter)
- **Recessive allele—**the allele encoded for but not displayed in a hybrid (lowercase letter)
- **Genotype—**the combination of alleles for a gene that an individual has
- **Homozygous dominant—**two dominant alleles
- **Heterozygous—**hybrid; one dominant allele, one recessive allele
- **Homozygous recessive—**two recessive alleles
- **Phenotype—**the physical expression of a genotype
- **Law of dominance—**a dominant allele will trump a recessive allele if an individual is heterozygous
- **Law of Segregation—**only one allele for a trait can be passed to offspring
- **Punnett Square—**a diagram for PREDICTING offspring genotype



- **Testcross—**crossing an individual with a dominant phenotype and an unknown genotype with a homozygous recessive individual to determine whether the dominant phenotype parent is homozygous dominant or heterozygous. If any children display the recessive phenotype, the dominant phenotype parent must be heterozygous.
- **Monohybrid cross—**crossing two heterozygous individuals for one trait (3:1 ratio)
- **Dihybrid cross—**crossing two individuals that are heterozygous for **two** traits (9:3:3:1 phenotype ratio)
- **Law of Independent Assortment—**alleles are inherited independently; one set of homologues lines up and separates independent from another

- **Non-Mendelian Genetics—**genetics that go beyond the basic principles of dominance and independent assortment that Mendel discovered
- **Complete Dominance—**in a heterozygous individual, the recessive allele is completely hidden

- **Incomplete Dominance**—in a heterozygous individual, the displayed phenotype is somewhere between the two phenotypes (dominant and recessive)
  - *** Example: the offspring of a red and white snapdragon will be a pink snapdragon
- **Codominance**—in a heterozygous individual, both alleles are equally dominant, meaning that both phenotypes are simultaneously displayed
  - *** Example: a white and a black chicken will produce a black-and-white speckled chicken
- **Multiple Alleles**—when a trait has more than two possible alleles (an individual can still only carry two alleles, but there is a wider range of alleles to carry)
  - *** Example: there are three alleles for blood type: I_A is equally as dominant as I_B, and both are completely dominant over i. I_AI_B is AB blood type, I_AI_A or I_Ai is A blood type, I_BI_B or I_Bi is B blood type, and ii is O blood type.
- **Polygenic Inheritance**—more than one gene encodes for a specific trait
  - *** Example: skin color, eye color
- **Environmental Impact/Epigenetics**—phenotype can be altered by environmental conditions
  - *** Example: hydrangeas grow blue in acidic soil, but pink in more basic soil. Similarly, rabbits or foxes will grow white coats during the winter regardless of their phenotype.
  - *** If two individuals have the same genotype but a different phenotype, than the environment must have had some impact.
- **Nonnuclear Inheritance**—mitochondrial or chloroplast DNA encodes for a trait. The egg is the only one that passes on mitochondria or chloroplasts to the zygote, meaning that it can only be passed down maternally.
- **Sex-Linked Genes**—genes that are encoded on sex chromosomes, meaning that an individual’s sex has bearing on their genotype.
- **X-Linked Recessive Trait**—males inherit the trait if the only X chromosomes they inherit has the trait, and females require two affected X chromosomes to inherit the trait. Affected chromosomes are written as X_{trait}.
- **X-Linked Dominant**—Both males and females inherit the trait if they inherit just one affected allele.
- **Carrier**—one who carries an allele for a trait but does not display the trait, especially in X-linked recessive cases.
- **Carrier Female**—a $X_{trait}X$ individual who does not display the trait but carries an allele (X-linked recessive).
- **Gene-linkage**—certain genes are close together on the same chromosomes, meaning they can be inherited together more often than not.
- **Recombinant**—crossing over occurs between two genes
- **Recombination Frequency**—total percentage of recombinant phenotypes produced between a dihybrid and an individual recessive for both traits
- **Map Units**—distance between genes, if higher, that means genes are farther apart on the same chromosome. 1 Map unit represents 1% recombination frequency.
- **Chi Square (see math note)**—can use to test if two genes are linked or not
- **Pleiotropy**—one gene controls multiple phenotypes
  - *** Example: sickle-cell anemia

- **Pedigree—**a family tree tracking the genotypes and phenotypes of different family members for a trait.
- **Nondisjunction—**errors during anaphase I or anaphase II in meiosis leads to aneuploidy and disorders. This can happen with both autosomal and sex chromosomes.
- **Aneuploidy—**the disorder of having an abnormal amount of chromosomes
  *** Example: Down Syndrome (Trisomy 21)
- **Aneuploid—**organism with an abnormal amount of chromosomes